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A Primer on US vs. EU Collateralized Loan Obligations (CLOs)

Collateralized Loan Obligations (CLOs) comprise an over \$1 trillion market that has spanned both sides of the Atlantic since before the Great Financial Crisis. US and European CLOs share many commonalities – at their core, both are securitizations of leveraged loans that provide the equity investor with non-recourse term financing without any requirements to post additional proceeds, along with the option to call or refinance.

But despite US and European CLOs both being built from the same skeleton, **there are important differences between the two markets and deal structures that we would like to highlight throughout this primer.** While we have previously discussed some of these differences in piecemeal, our intent is for this report to act as a singular comprehensive reference point for those looking to dive into the differences between US and EU CLOs. To that end, we cover a whole range of topics from collateral and structural differences to market dynamics and documentation language differences.

This primer is meant to act as an accompaniment to the work that we have already done in our other primers, namely our [primer on CLOs](#) and our [primer on Private Credit CLOs](#). As such, to avoid redundancy, we will not be going in-depth on many CLO concepts and would suggest that those looking for a refresher refer to those reports.

One important note: As of the date of publication, while there is a healthy private credit CLO market in the US, the market is yet to take off in Europe, so the comparisons in this primer are between each market's broadly syndicated loan (BSL) CLO transactions, unless otherwise specified.

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All figures as of August 2024 unless otherwise indicated.

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Summary of Differences between US and EU BSL CLOs

	Feature	US CLOs	EU CLOs
Market	BSL market size	\$905 bn	€235 bn
	Typical annual new issuance	~\$100-\$200 bn	~€20-40 bn
	Number of managers	156 managers	67 managers
Collateral	Diversification	More diverse. Median Moody's Diversity Score: 83	Median Moody's Diversity Score: 60
	Country mix	Vast majority US	Mixed, with France, UK and US based issuers as the top 3
	Loan vs. bond & fixed vs. floating	Up to 5% bond exposure to be Volcker compliant. Small fixed rate exposures.	30% limit on bonds (about 14% bond exposure currently). More fixed rate collateral.
	CCC and below exposure	US CLOs typically have higher CCCs	Fewer CCCs, but more privately rated loans, which show up as NR
Structure	Debt-to-equity	Deals typically consist of ~8% equity to ~92% debt	Similarly typically consist of ~8% equity to ~92% debt
	Debt stack	Typically AAA to BB	Typically AAA to B
	Credit enhancement (CE)	Typical AAA CE at 36-38%	Typical AAA CE at 39-40%
	Tranche currency/type	USD, floating linked to SOFR	Euro, floating linked to Euribor
Regulations	Risk Retention (RR)	Risk retention rules repealed in 2018, some deals still elect to satisfy RR rules to access an expanded investor base	5% risk retention by vertical or horizontal slice. All EU CLOs are RR compliant.
ESG	Negative screening	ESG exclusions are growing but well behind the EU	Industry exclusion is more or less a market standard in EU since 2021
	In-house ESG scoring	Not yet introduced in the US CLO market	A few managers also score loans and report ESG score of their CLOs

Source: Intex, Pitchbook | LCD, Dealscribe, Morgan Stanley Research

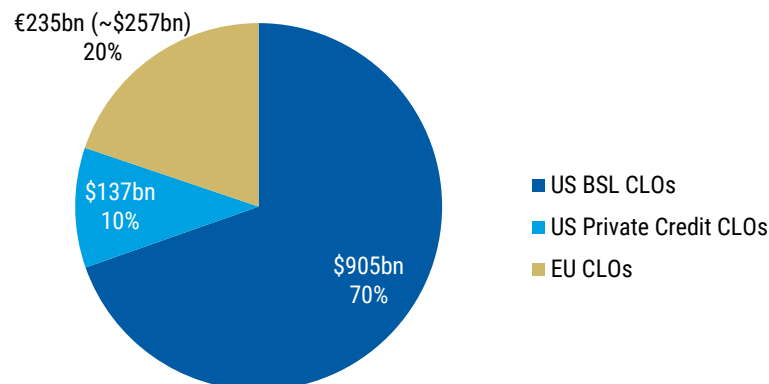
SECTION 1

Market Overview

Comparing US and EU CLO Market Sizes

- **US vs. European CLOs:** The CLO market has a significant presence across the Atlantic, with a total of about \$1.3 trillion currently outstanding, of which almost \$1.2 trillion is made of BSL CLOs.
- Typically, a USD-denominated CLO is issued by a US-based manager and primarily acquires loans of US-based companies. Euro-denominated CLOs are similarly issued by an EU-based manager, but their portfolios include issuers from the euro area, the UK, and the US.

Global Outstanding CLOs by Region



Source: Intex, Morgan Stanley Research. Size is based on deal size which may differ from the sum of tranche balances.

Sizing the BSL CLO Market by Rating and Tenor

Rating	US CLO Universe (\$bn)			EUR CLO Universe (€bn)		
	Short RP ≤ 2026	Long RP > 2026	Total	Short RP ≤ 2026	Long RP > 2026	Total
AAA	\$325	\$225	\$550	€ 100	€ 37	€ 137
AA	\$75	\$42	\$117	€ 19	€ 7	€ 26
A	\$39	\$21	\$60	€ 12	€ 4	€ 16
BBB	\$38	\$21	\$59	€ 11	€ 4	€ 15
BB	\$28	\$12	\$40	€ 10	€ 3	€ 13
B	\$3	\$1	\$3	€ 5	€ 2	€ 7
Equity	\$90	\$36	\$126	€ 19	€ 5	€ 25
Total	\$596	\$359	\$955	€ 177	€ 61	€ 238

Source: Intex, Morgan Stanley Research. Short deals are those with reinvesting periods ending in 2026 or earlier. Does not include private credit or middle market deals.

Global Issuance by Vintage



Source: Intex, Morgan Stanley Research. Issuance year is based on closing date and includes Private Credit CLOs.

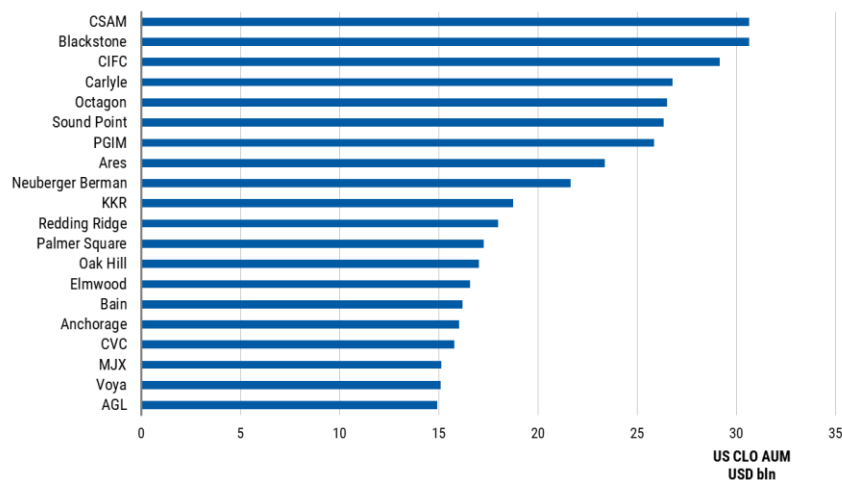
CLO Managers on Both Sides of the Atlantic

- The manager base is understandably bigger in the US compared to Europe, with 150+ active managers in the BSL space in the US compared with 65+ in EU.
- There is, however, much overlap among the CLO managers that operate in the US and EU markets. About 30% of US managers are also active in Europe, while 70% of EU managers are active in the US.

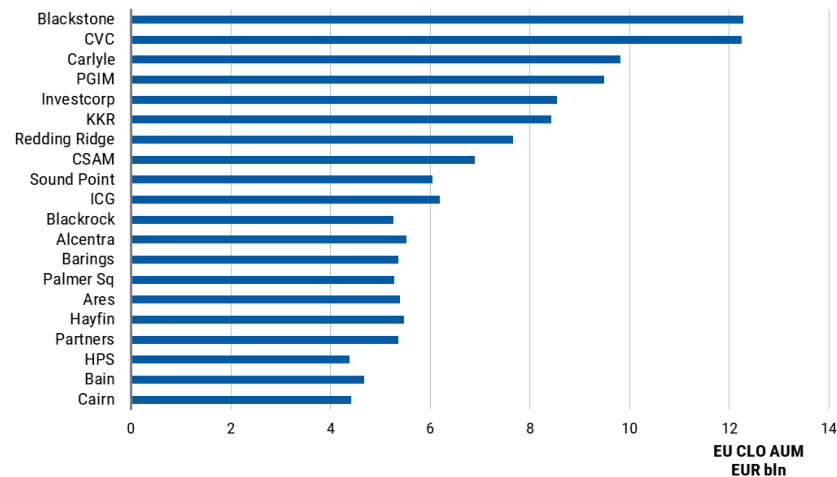
Manager Landscape

CLO Manager Landscape	US	EU
Manager Count	156	67
Largest Manager by AUM	CSAM	Blackstone
AUM of Largest Manager	USD 30.7bln	EUR 12.3bln
Top 5 Managers by AUM	CSAM, Blackstone, CIBC, Carlyle, Octagon	Blackstone, CVC, Carlyle, PGIM, Investcorp
Share of Top 5 Managers	15.3%	22.0%

Top 20 US Managers by AUM



Top 20 EU Managers by AUM



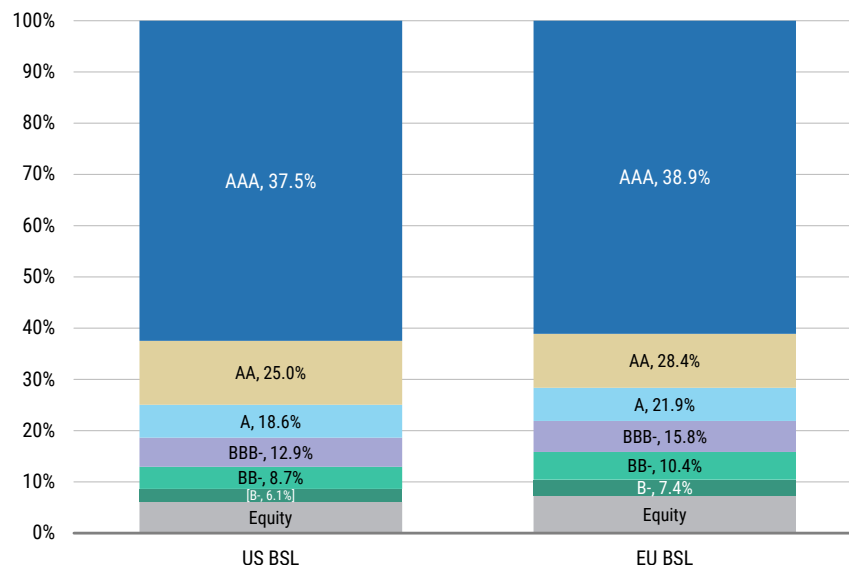
SECTION 2

Structure Comparison

Structure Comparison

- Both US and EU CLOs are structured with a range of debt tranches and an equity tranche at the bottom. However, we wish to highlight:
- NC/RP (Non-Call Period/Reinvestment Period):** The most common format in US CLOs is 2yr NC/5yr RP compared to 1.5yr NC/4.5yr RP for Europe. A fair number of EU CLOs also have a 2yr NC, but over the past 3-4 years 1.5yr NC has been more popular in the EU.
- Static CLO Universe:** Static CLOs are a small part of both markets, making up ~2% of each market by deal count.
- Subordination:** US CLOs have lower par subs and are thus slightly more levered on a tranche-by-tranche level; however, debt-to-equity leverage is comparable due to inclusion of single-B tranches in the EU.
- Single-B Tranches:** EU CLOs routinely include a single-B tranche in their debt structure. In the US, they are significantly less common (only \$3bn of single-B and \$40bn of BBs outstanding, compared with €7bn and €13bn in the EU, respectively).
- Fixed-Rate Tranches:** Both US and EU CLO tranches are primarily floating-rate, linked to SOFR in the US and Euribor in the EU. Sometimes deals on both sides of the Atlantic will include minimal fixed-rate tranches (for example, there are a few AA tranches offered in a fixed format in the EU).

US vs. EU Weighted Average Par Subordinations by Tranche



Source: Intex, Morgan Stanley Research

SECTION 3

Collateral Comparison

Collateral Limits

- **The asset composition of CLOs in both markets is broadly similar:** They are mostly made up of 1st lien, senior secured, floating-rate leveraged loans. However, there are a few key differences in the limitations on what a CLO manager can buy – some of which stem from the different regulatory regimes.
- One key area where the collateral concentration limits differ is **bond** exposures. The Volcker Rule in the US limits bond exposures to 5% (the occasional deal will still include a higher limit) while EU CLOs can generally have up to 30% in bonds (due to the relative lack of loan supply). Along similar lines, **fixed-rate** exposures of up to 10% are allowed in Europe versus 5% in the US.
- Other differences are mainly related to the **diversity** of collateral, where a smaller and more concentrated loan market in Europe leads to slightly more lenient limits on obligor and industry concentrations.

Typical Concentration Limits in US and EU CLOs

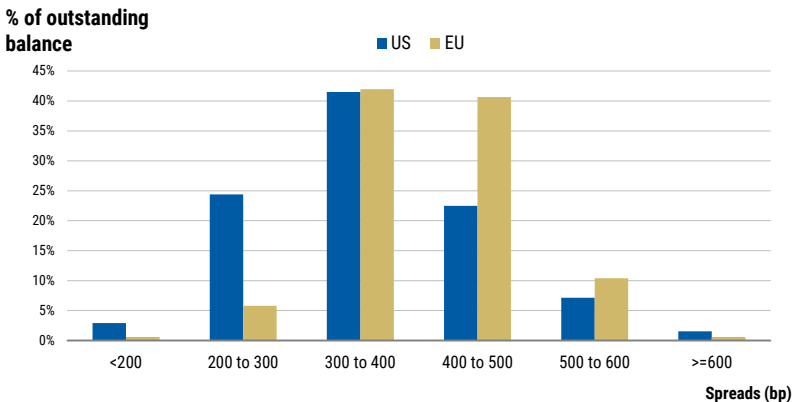
Typical Concentration Limits	US	EU
2nd Lien and Unsecured	10.0%	10.0%
Bonds	5.0%	30.0%
Fixed Rate	5.0%	10.0%
Single Obligor	2.0%	2.5%
Largest 10 Obligators	22.5%	26.5%
Single S&P Industry	15.0%	17.5%
DIP Loans	7.5%	5.0%
Cov Lite*	60.0%	30.0%
CCC Limit	7.5%	7.5%
Minimum Obligor Size	250m	250m

Source: Dealscribe, Morgan Stanley Research. Note: The limit for covenant-lite excludes collateral where related assets contain covenants.

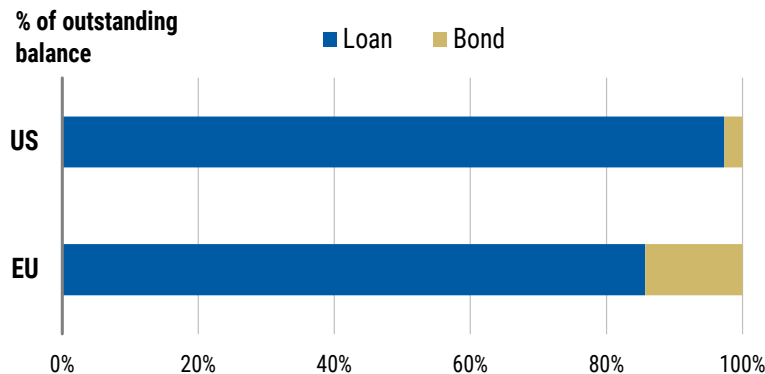
Collateral Composition Comparison

- **Spread Distribution:** European leveraged loans typically have higher spreads, averaging around 400bp over Euribor, compared to about 350bp over SOFR in the US.
- **Bonds/Fixed-Rate:** In line with larger limits, EU CLOs have many more bonds (14% compared to 3% in the US) and fixed-rate assets (8% compared to 3% in the US). Most of these bonds in EU CLOs are senior secured.

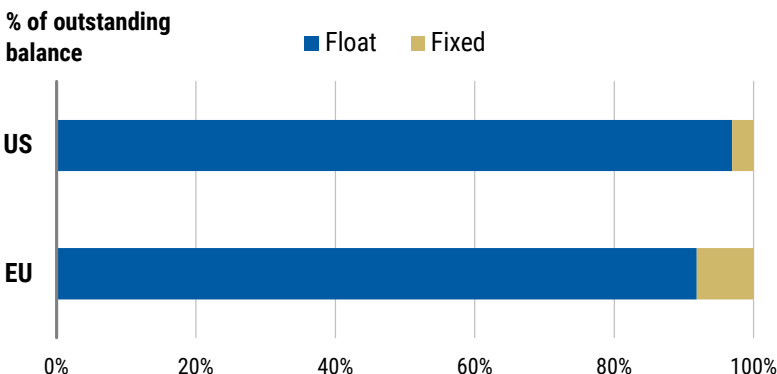
Loan Spread Distribution



Loan vs. Bond Split



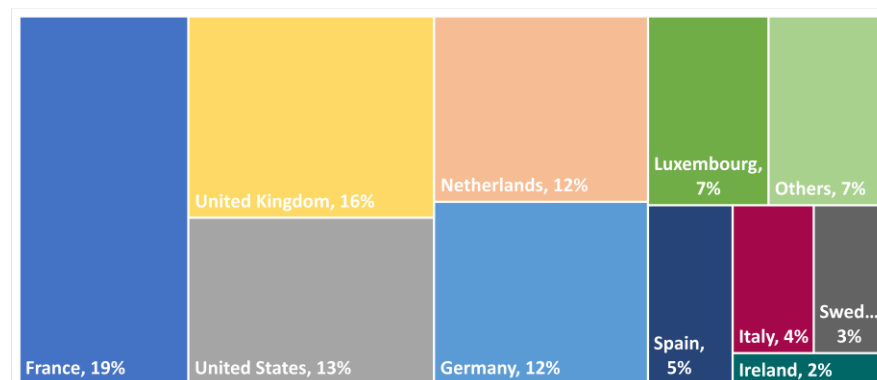
Float vs. Fixed Split



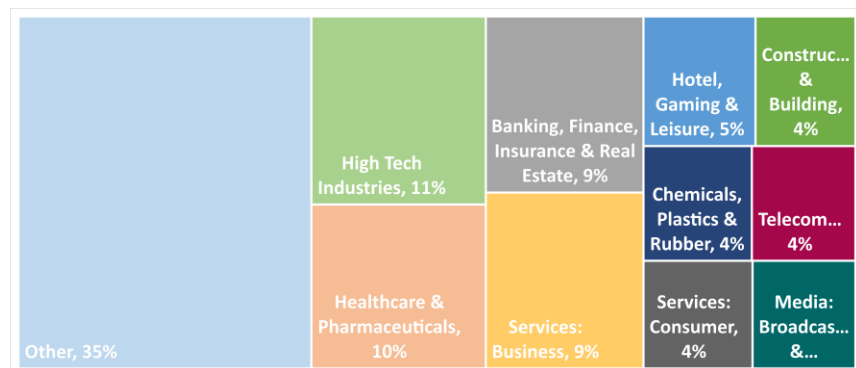
Collateral Composition Comparison (Continued)

- **Country Composition:** While US CLOs do have buckets for collateral originated outside the US, the vast majority of US CLO collateral is from the US. By contrast, EU CLOs include collateral from a multitude of countries, with the most common country (France) at ~20% of the collateral.
- **Industry Composition:** Both US and EU CLOs have limits on industry concentrations; as such, no one industry dominates either market. Currently, the largest industries in the US are Tech and Healthcare, compared to Healthcare and Business Services in the EU.

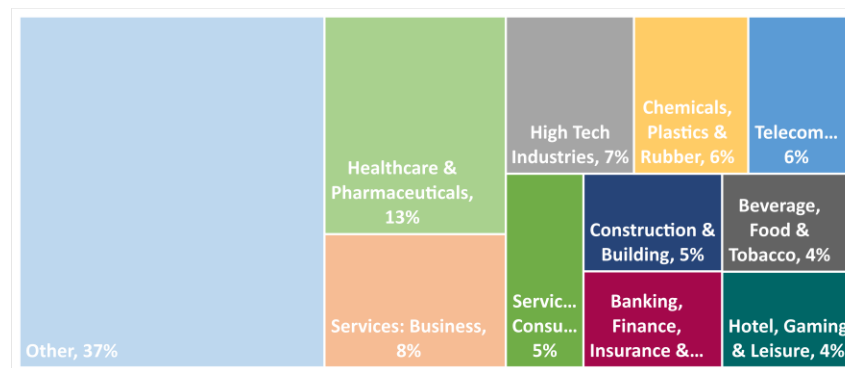
EU Collateral Country Split



US Collateral Industry Split



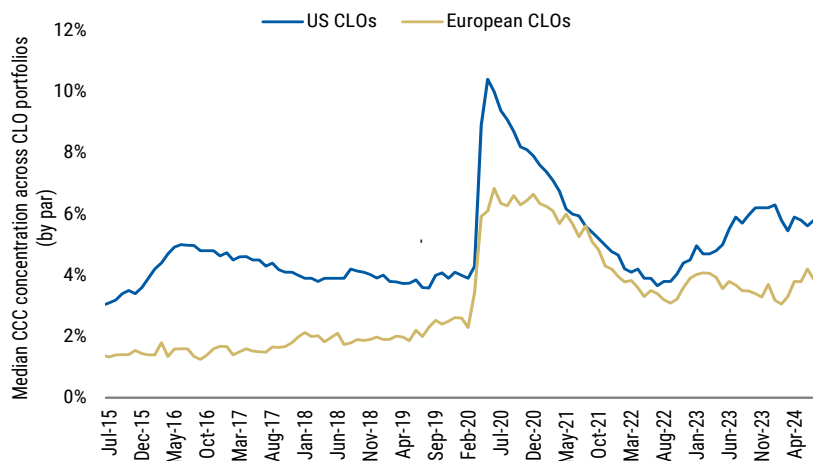
EU Collateral Industry Split



Collateral Composition Comparison (Continued)

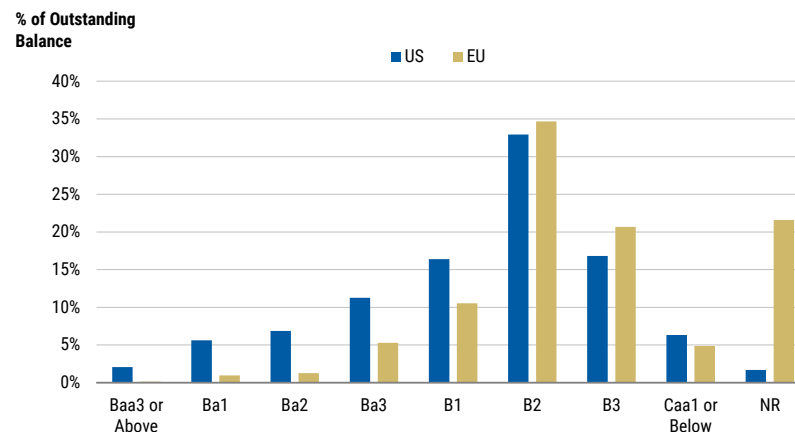
- **Ratings Composition:** Since the underlying in both markets are leveraged loans (and HY bonds), the ratings mix of the collateral is rather similar, with the caveat that EU CLOs have a significantly larger share of privately rated loans – these show up as NR in trustee reports, but the correct private rating still applies (for CCC tests, for example).
- Single-B rated collateral dominate both US and EU CLOs, comprising over half of the share by outstanding balance. US CLOs have a larger proportion of double-B rated collateral compared to EU CLOs. At the same time, the share of CCC assets is higher in the US (ignoring the aforementioned privately rated assets).

Median CCC Concentrations

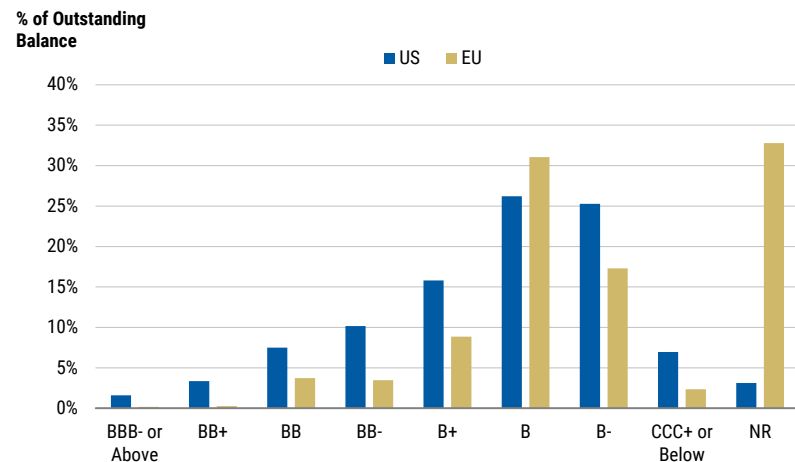


Sources for this slide: Intex, Morgan Stanley Research

Split by Rating – Moody's



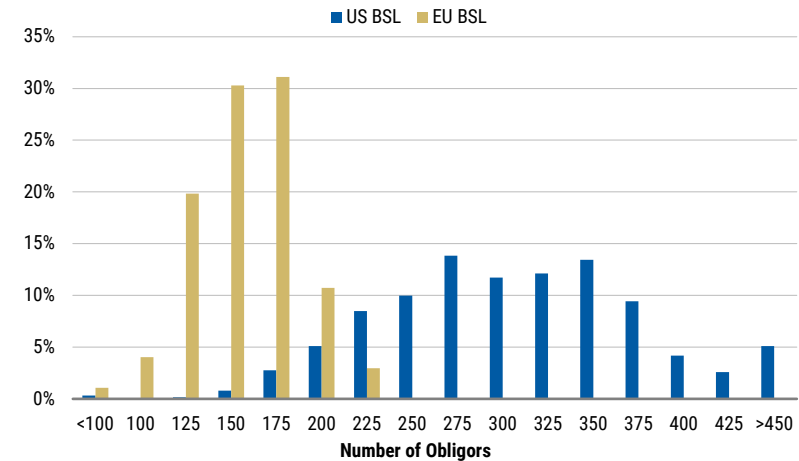
Split by Rating – S&P



Collateral Composition Comparison (Continued)

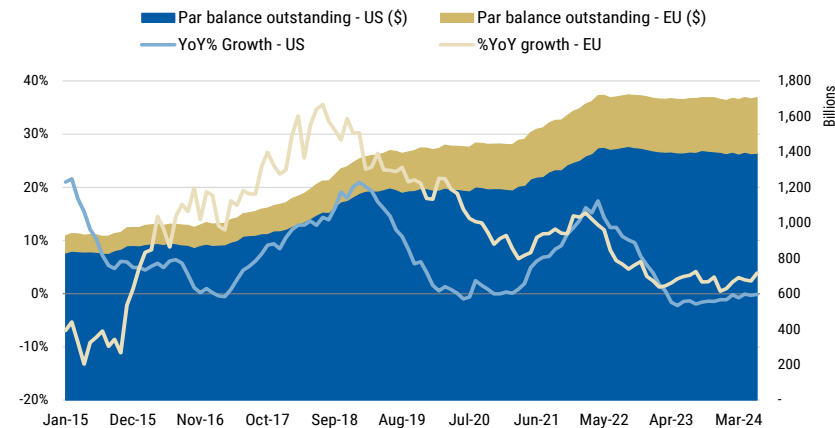
- **Obligor Count:** One important difference is that US CLOs tend to have significantly more obligors compared to EU CLOs. Most US deals have between 250 and 350 obligors, compared with 125 to 200 in the EU.
- **Loan Balance:** The loan and CLO markets on both sides of the pond have grown significantly since 2015, although that growth has slowed down since 2023. BSL CLO balances as a percentage of loans have increased in recent years, with US CLOs at around 70% and EU CLOs at around 80% of their respective loan markets.

Distinct Obligor Count



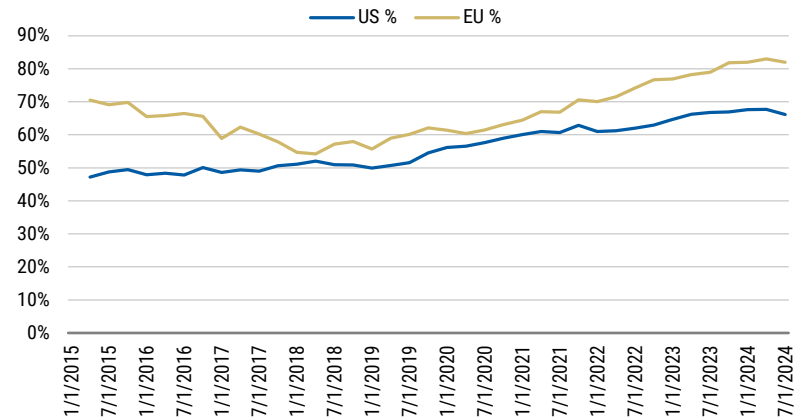
Source: Intex, Morgan Stanley Research

Outstanding Loan Index Par Balance



Source: Pitchbook | LCD, Morgan Stanley Research

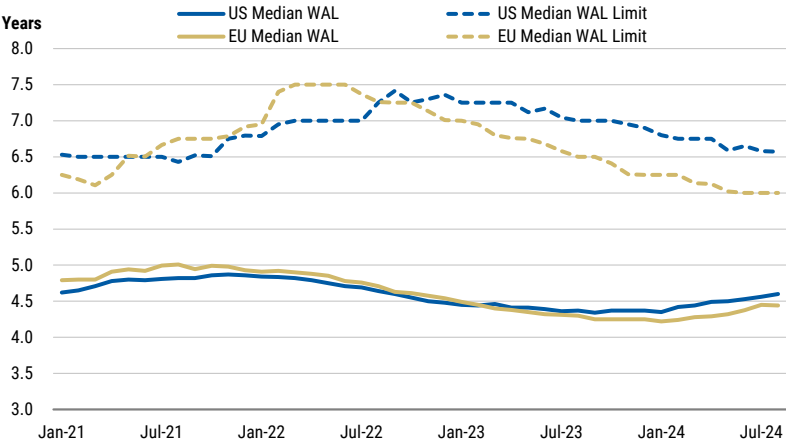
BSL CLO Balance as a Percentage of Loan Balances



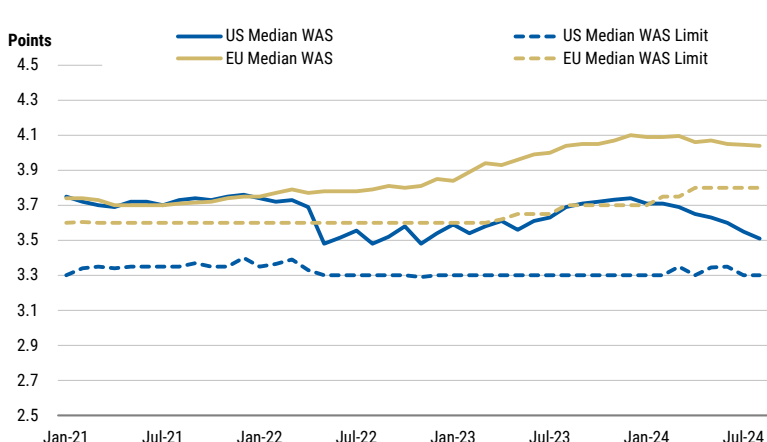
Source: Pitchbook | LCD, Intex, Morgan Stanley Research

Collateral Quality Tests

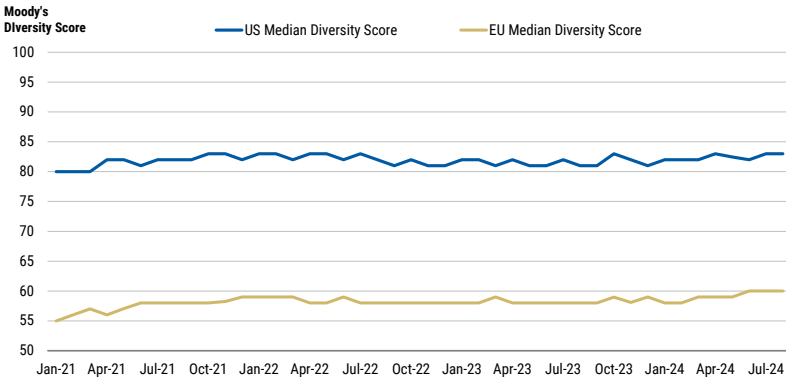
Weighted Average Life (WAL) Test



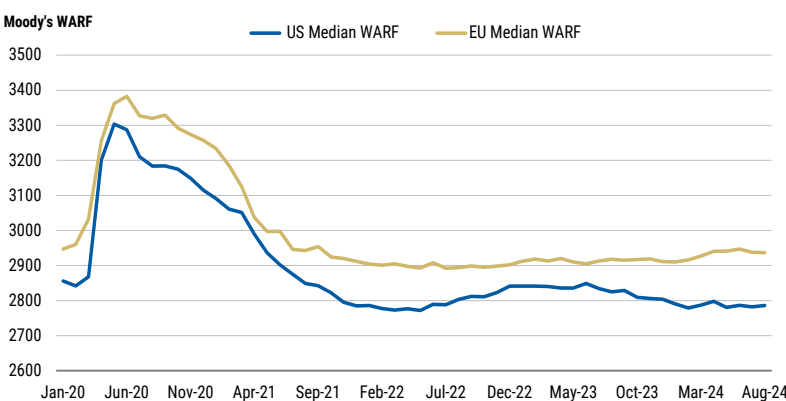
Weighted Average Spread (WAS) Test



Diversity Score

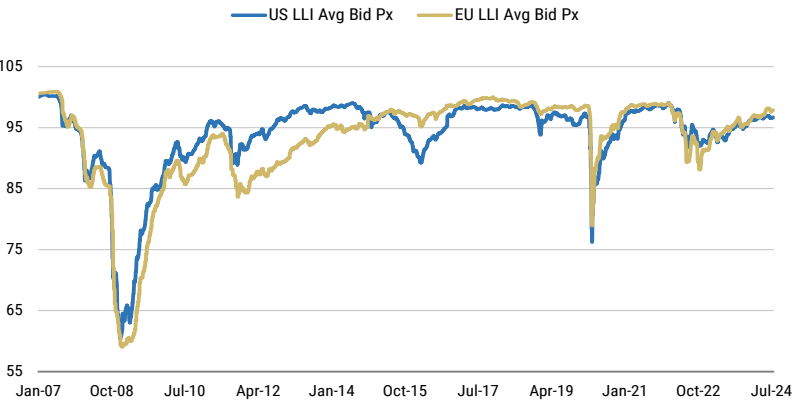


Weighted Average Rating Factor (WARF)



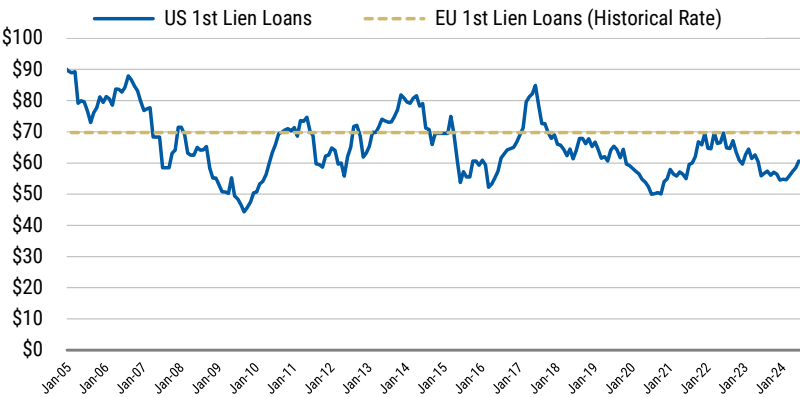
Collateral Performance Comparison

Loan Index Weighted Average Bid Price



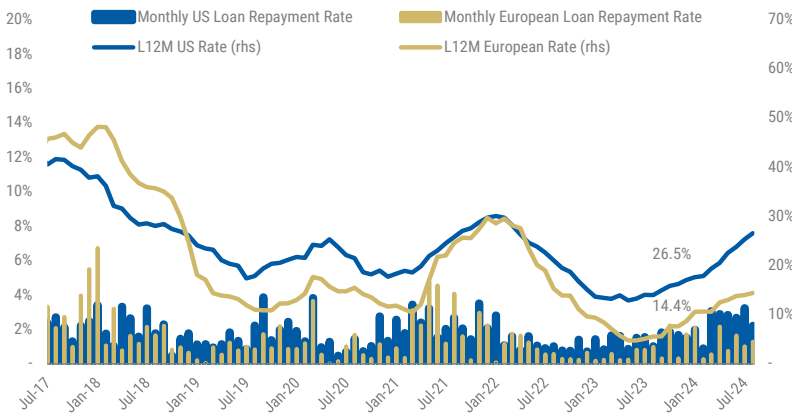
Source: Pitchbook | LCD, Morgan Stanley Research

Loan Recoveries



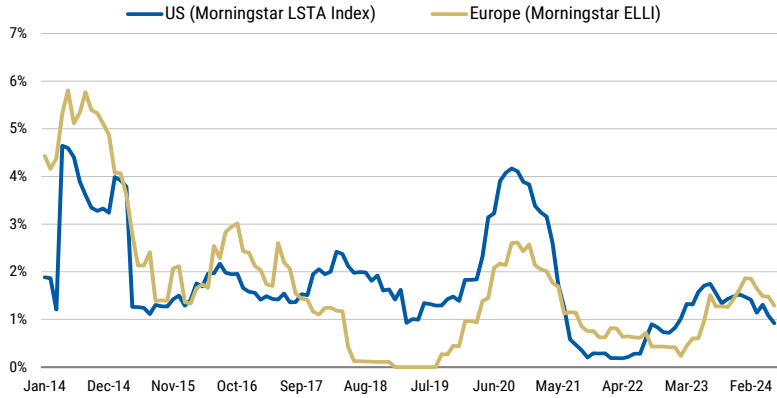
Source: Pitchbook | LCD, Moody's, Morgan Stanley Research. Moody's EU Loan recovery data is a historical estimate from 2017, included for comparison.

Loan Index Monthly Repayment Rate



Source: Pitchbook | LCD, Morgan Stanley Research

Loan Index LTM Default Rate



Source: Pitchbook | LCD, Morgan Stanley Research

SECTION 4

Documentation Comparison

Post Reinvestment Language

- US and EU CLOs differ slightly on typical post-reinvest language in their documents. The key area where the difference matters the most is the **WAL Test**, which is generally more restrictive for EU CLOs. A majority of EU CLOs have a one-touch test, which punishes* the deal if it is not passing its WAL test at the time of exiting the RP.
- By contrast, US CLOs typically require “maintain or improve” independent of whether the WAL test had been passing at RP exit.
- In addition, EU CLOs typically require more collateral quality test checks.
- In the exhibit below, we show “typical” post-reinvest language across US and EU CLOs, noting that these terms can vary.

Typical Post Reinvestment Language	US CLOs	EU CLOs
WAL Test	Maintain or improve	One touch*
OC Test	Pass	Pass, more EU CLOs require passing before and after each reinvestment
Other Collateral Quality Tests	Usually just IC, some CLOs also require OC and WAL	Usually IC, CCC and WARF, some CLOs need IC and CCC only
Maturity of Replacement	Earlier or equal to replaced collateral	Earlier or equal to replaced collateral
What can be reinvested?	Unscheduled principal payments, sales of credit risk obligations	Unscheduled principal payments, sales of credit risk obligations, sales of credit improved obligations
Trading Plans - Difference in Shortest and Longest	Typical max difference between longest and shortest asset can be 3 years	Typical max difference between longest and shortest asset can be 3 years
Trading Plans - Minimum Maturity for Shortest Collateral	Minimum maturity of shortest asset can be 6 months	Minimum maturity of shortest asset can be 6 months

*Note: For “one-touch” WAL tests, if the CLO is failing its WAL test at the end of the reinvestment period, then any post-RP reinvestments require that the WAL test be satisfied. If the CLO is instead passing its WAL test, subsequent reinvestments are typically allowed to fail the test so long as the test is improved.

Source: Dealscribe, Morgan Stanley Research

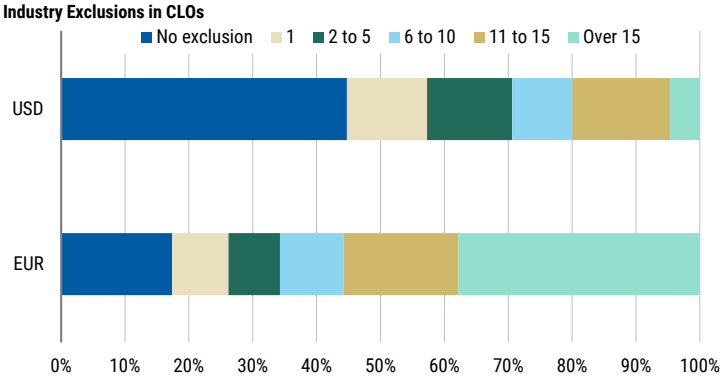
Additional Documentation Differences

- **Par Flush and Trading Gain Payouts:** Most US and EU CLOs have a Par Flush mechanism where in the early (typically the first or second) payment dates, par gains can be flushed out to equity. Both US and EU CLOs typically have a cumulative limit of up to 1% of deal par that is allowed to be flushed out.
- Most EU CLOs have an additional **trading gains** payout mechanism. The payout of trading gains (again usually limited to a cumulative 1% of par) is allowed in most EU CLOs at any time, rather than just at the early payment dates. This feature is not common in US CLOs.
- **Turbo Tranches:** A very small handful of EU CLOs have turbo single-B tranches where a portion of the excess interest that would normally go to the equity (usually 20%) is instead used to pay down the turbo debt tranche. This helps to sell the junior mezz tranche at tighter levels; plus, once the principal on the turbo tranche is paid down, the weighted average cost of debt financing is lowered for the equity.
- In the US and EU, there are occasionally additional tranches senior to the rest of the debt (usually denoted as Class X) that amortize using interest payments on a set schedule at the beginning of the deal. These are sometimes included in resets to reduce any required equity injections.

ESG Language in CLO Documents

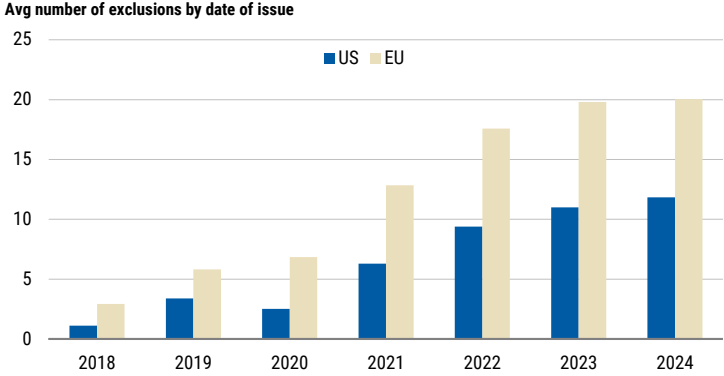
- With respect to ESG, the EU CLO market is a step ahead of the US, with negative screening (exclusion of certain industries) being almost a market standard now. Well over 80% of the EU CLO market has some form of exclusion, compared to 55% of US CLOs.
- The list of excluded industries has also grown over the past few years in both markets. The average EU CLO now excludes about 20 industries, compared to roughly 11 for the US.
- A handful of EU CLO managers also score and rate collateral on ESG considerations based on their in-house frameworks. They have target ESG scores for their CLOs, which they report on a regular basis. So far, US CLOs have only implemented negative screening.
- Within both EU and US CLOs, controversial weapons, tobacco, and thermal coal are the three industries excluded in most CLOs. Certain industries like oil and gas, midstream oil sands, and multilateral organization guidelines (UNGC, etc.) find more mentions in EU CLOs versus US CLOs.

ESG in EU is a Step Ahead of the US



Source: Dealscribe, Morgan Stanley Research. Note: Only CLOs issued/reset since 2017

Number of Excluded Industries is Up in Both Markets



Source: Dealscribe, Morgan Stanley Research. Note: Only CLOs with exclusions.

European Securitization Regulation

Securitization Regulation

- The all-encompassing Securitization Regulation (SR) came into force in Jan-2019.
- The SR lays out due diligence (Art 5), retention (Art 6), and transparency (Art 7) requirements for all market participants. It also lays out the STS (Simple, Transparent, Standardized) framework and the capital requirements.
- Leveraged loan CLOs fall under the non-STs category owing to their actively managed nature.

Risk Retention (Article 6)

- The risk retention requirement remains in place for all securitizations, and the new regulation also places the onus of retention on the originator, sponsor, or original lender.
- The requirement is a minimum retained interest of 5% of the nominal value of the transaction for the life of the transaction
- Even if the investor does not need risk retention (non-EU investor) the EU originator / sponsor / lender still needs to retain the 5% “skin in the game.”
- All European CLOs are risk-retention compliant.

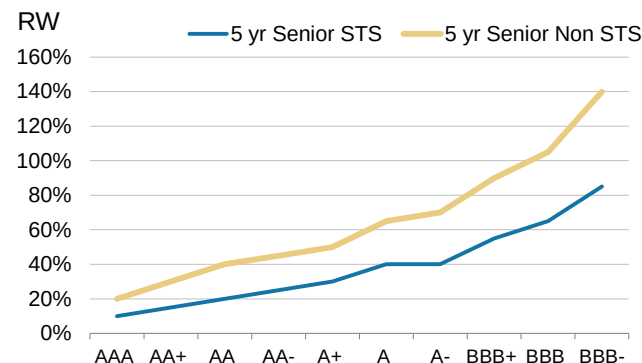
Capital Requirements for Bank Investors

- There are three hierarchies for capital calculation: SEC-IRBA (Internal Ratings Based), SEC-SA (Standardized Approach) and SEC-ERBA (External Ratings Based).
- For non-STs (CLOs), the risk weight floor is 15% under all the hierarchies.
- Under SEC-ERBA, risk weights for the stack depend on the rating and the WAL.

Capital Requirements for Insurer Investors

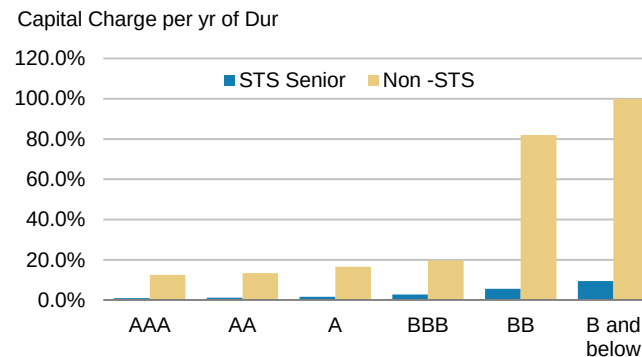
- The STS framework is now also a part of the Solvency 2 capital framework.
- CLOs being non-STs attract a higher capital charge under the regulation, and the tiering between STS and non-STs is rather wide.
- At the AAA level, CLOs are subject to a capital charge of 12.5% per year of duration under the standard approach.

Risk Weights for Bank Investors



Source: Morgan Stanley Research, European Commission. Note: Exhibit shows only IG tranches and shows RW under the External Ratings Based Approach.

Capital Charges for Insurers



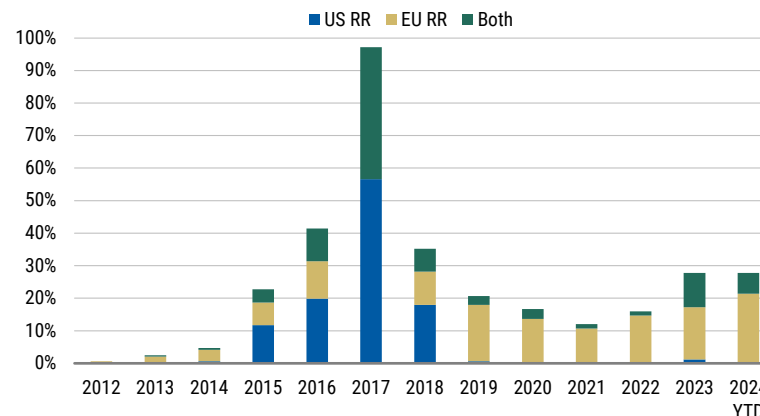
Source: Morgan Stanley Research, European Commission

US Securitization Regulation

Risk-Retention Rules

- The requirement for US CLOs to be risk-retention compliant was repealed in 2018
- Since then, a number of deals have voluntarily elected to comply with risk-retention rules to access a wider base of investors, especially investors outside the US
- In recent years, roughly 15-25% of US new-issue deals have satisfied risk retention in some form (EU RR, US RR, or both)
- US risk retention was initially adopted at the end of 2016 – there are a few distinctions between US and EU risk retention (e.g., US RR measures retention requirements by fair value, as opposed to nominal value in EU RR), but the overall intent (having “skin in the game”) is similar

US New Issue CLO Risk Retention Compliance



Source: Pitchbook | LCD, Morgan Stanley Research. Data as of August 26th 2024.

US Bank Demand, RWAs and Basel 3

- US banks are a key investor for the senior portions of the US CLO capital structure
- While an oversimplification, bank demand for securities can be thought of as constrained by some combination of risk-weighted assets (RWAs), high-quality liquid assets (HQLAs), and duration
- Currently, CLO tranche RWAs are calculated using the SSFA framework, which relies heavily on credit enhancement, subject to a floor of 20% (for context, this is in line with GSE-guaranteed mortgages)
- The current Basel 3 proposal lowers that floor to 15%, potentially making CLO AAAs more attractive than agency MBS from an RWA perspective
- CLO tranches are not treated as HQLAs, and as such are not eligible assets for the liquidity coverage ratio

Insurance Company Demand and the NAIC

- Insurance companies are an important source of demand for the IG portions of CLO tranches
- The NAIC uses ratings to assign CLO debt to one of 19 different risk buckets, which correspond to risk-based capital (RBC) factors
- In August 2024, the NAIC approved the adoption of a 45% capital charge for residual tranches (i.e., CLO equity)
- The NAIC intends to move away from a ratings-based methodology to a modeling framework, but current implementation has been delayed to year-end 2025

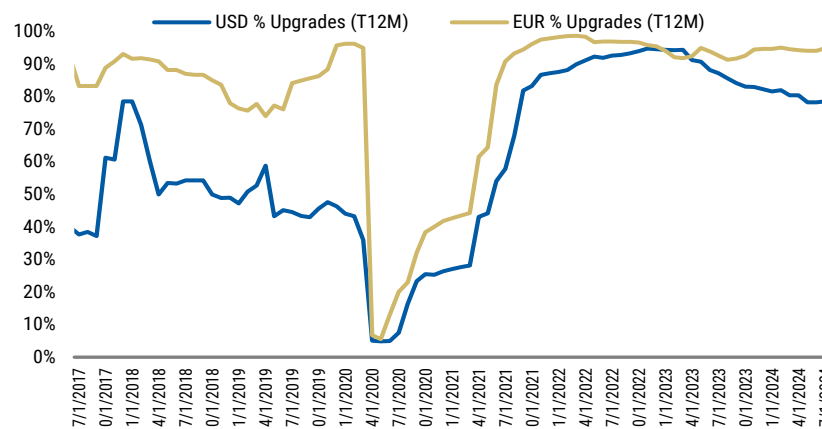
SECTION 5

Performance Comparison

Downgrades and Defaults

- **Tranche rating change** activity has historically been small, up until we saw an immense amount of downgrades in both markets in 2020, followed by mostly upgrades in the years since.
- **Defaults or principal write-downs** on CLO tranches have been few and far between across both US and EU CLOs of the pre-GFC era. US CLOs have had a slightly worse outcome, with one single-A rated bond having taken a write-down and more BBB bonds having been impacted compared to their European counterparts.
- No post-GFC CLO in the EU has taken a write-down, while 12 US 2.0 CLO tranches according to Moody's, along with 20 according to S&P, have defaulted so far.

Proportion of CLO Upgrades vs. Downgrades



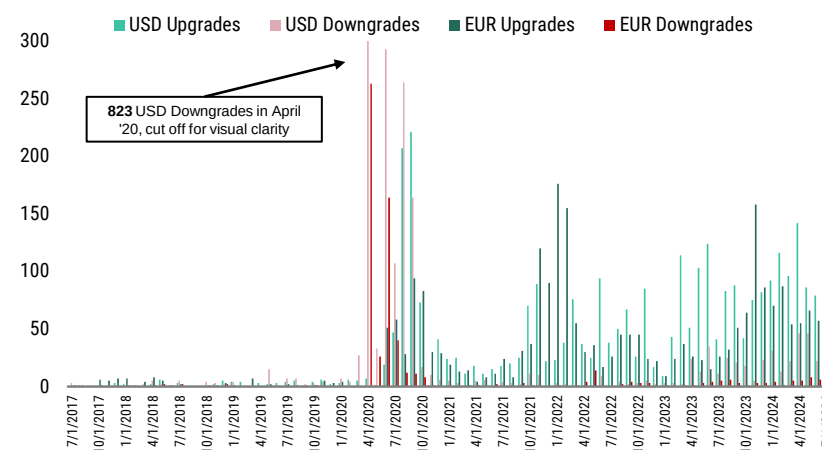
Source: Bloomberg, Morgan Stanley Research

Principal Impairments in CLOs Have Been Very Limited

US	Original Rating	Initial Size	Principal Impairments	Impairment Rate
	Aaa	5628	0	0.0%
	Aa	2668	0	0.0%
	A	2384	1	0.0%
	Baa	2266	23	1.0%
	Ba	2027	36	1.8%
	B	496	19	3.8%
	Total	15469	79	0.5%
Europe	Original Rating	Initial Size	Principal Impairments	Impairment Rate
	Aaa	1543	0	0.0%
	Aa	1427	0	0.0%
	A	962	0	0.0%
	Baa	890	3	0.3%
	Ba	779	15	1.9%
	B	578	0	0.0%
	Total	6179	18	0.3%

Source: Moody's, Morgan Stanley Research. Data from 1993 to 2023.

CLO Monthly Upgrades and Downgrades

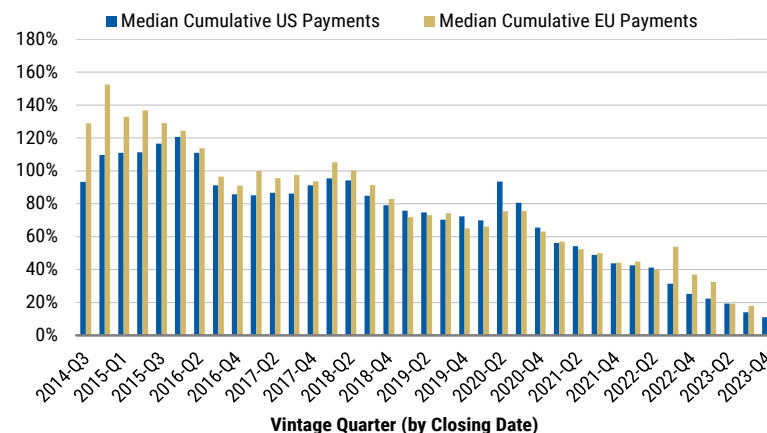


Source: Bloomberg, Morgan Stanley Research

CLO Equity and Overcollateralization Comparison

- **Cumulative equity distributions** (as a percentage of par) among the two markets generally track closely to each other, though EU deals from 2014-2015 had higher total payments.
- **Equity NAVs** between the two markets have also tracked relatively closely, although US NAVs tend to be slightly higher.
- **Junior overcollateralization (Jr. OC) cushions** in older vintage EU CLOs are higher, while more recent vintages have similar US and EU cushions.

Median Cumulative Equity Distributions by Vintage

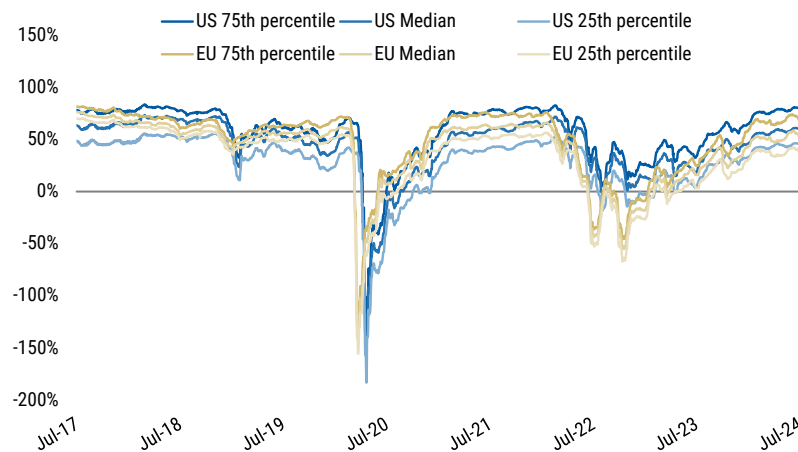


Median Jr. OC Cushions by Issuance Vintage

	Vintage	US CLOs		European CLOs	
		Jul-24	Jul-23	Jul-24	Jul-23
Median Junior OC Cushion	2013	3.3%	2.8%	-	-
	2014	1.9%	1.2%	-	-
	2015	3.4%	2.8%	4.1%	4.3%
	2016	3.4%	3.0%	4.2%	4.5%
	2017	3.5%	4.0%	3.7%	4.3%
	2018	4.0%	2.1%	4.3%	3.4%
	2019	3.5%	3.9%	4.5%	4.1%
	2020	4.6%	5.0%	3.9%	4.3%
	2021	4.4%	5.0%	4.4%	4.5%
	2022	4.8%	5.0%	4.7%	4.7%

Source: Morgan Stanley Research, Intex

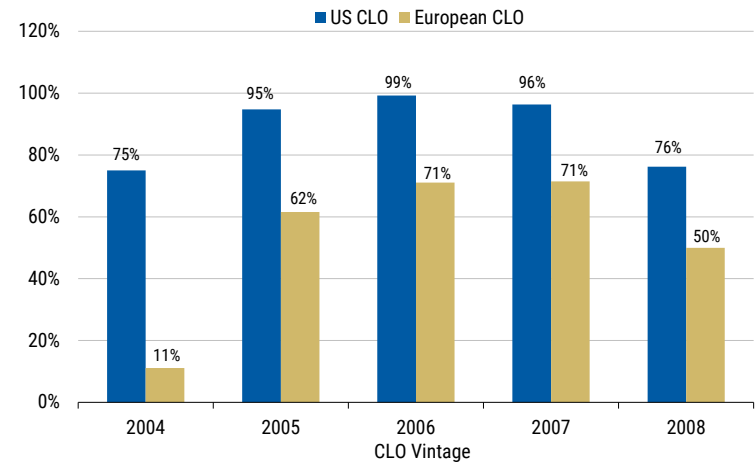
Equity NAVs



CLO GFC Performance Comparison

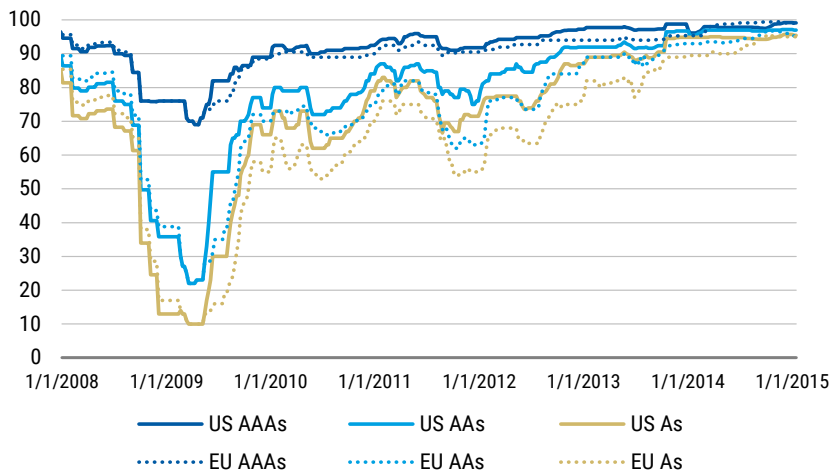
- **CLO 1.0 (pre-crisis) debt tranche prices:** Like most financial asset classes, the CLO market had a sharp sell-off during the 2008-09 financial crisis. Together with leveraged loans, the CLO tranches bottomed during the crisis but managed to recover fairly quickly (the recovery started in late 2009). Among all CLO tranches, the senior tranches recovered the fastest, followed by various mezzanine tranches, according to their seniority.
- **CLO 1.0 mezz performance vs. loans during the crisis:** CLO mezzanine tranches lost more in mark-to-market terms than loans entering the last default cycle, but the degree of price recovery in CLO junior mezzanine tranches out of the credit cycle was also asymmetrically higher than that of loans. See [Revisiting Lessons from the Financial Crisis \(26 Feb 2016\)](#) for more details.

Percentage of Pre-GFC Equity that Returned at Least Par



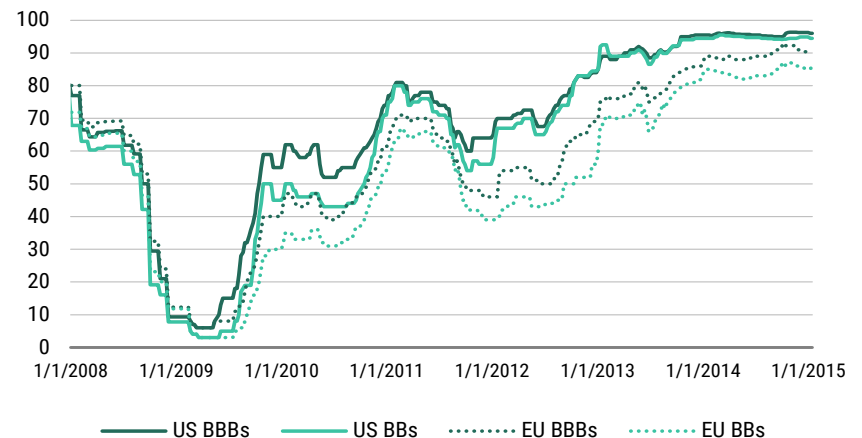
Source: Morgan Stanley Research, Intex

CLO AAA, AA, A Prices through the GFC



Source: Morgan Stanley Research

CLO BBB, BB Prices through the GFC

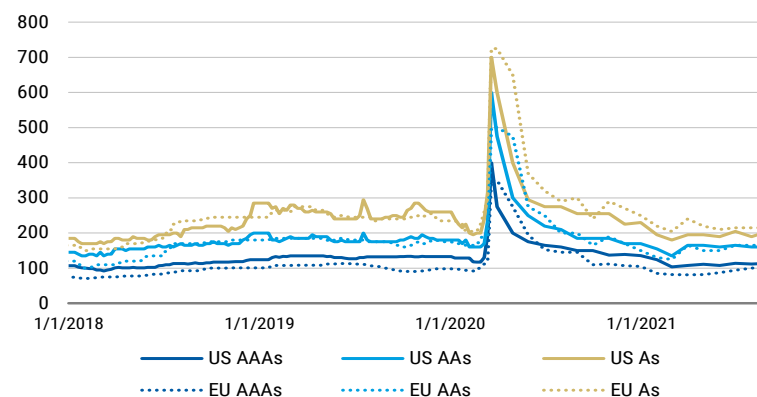


Source: Morgan Stanley Research

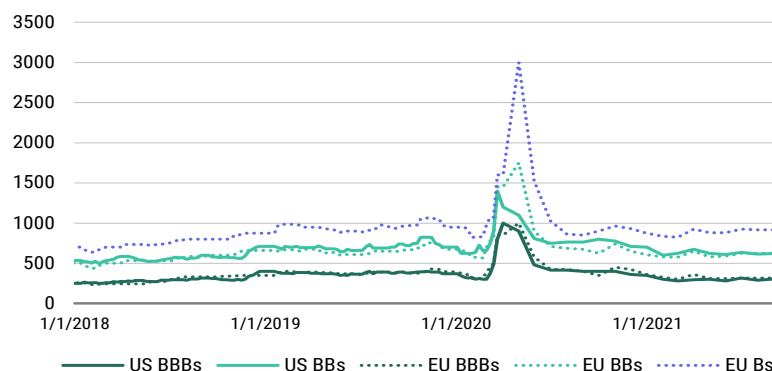
CLO COVID Performance Comparison

- **Loan Prices:** US leveraged loan prices began falling in February 2020 and hit a low on March 23, 2020. Prices began to recover quickly over the following two weeks, then after a few stalls recovered fully to pre-COVID prices by January 2021.
- **CLO Debt Prices:** The movement of CLO prices during COVID was similar to the movement during the GFC. There was a sharp sell-off in early to mid-March 2020, followed by a quick recovery. Once again, AAA led the recovery. As realized loan default levels remained lower than expected, more credit-risk-exposed tranches followed in their recoveries throughout the following year.
- **Tranche Downgrades:** The rolling-12-month downgrade percentage for loans in the US in December 2020 reached 48.25%, but CLO tranche downgrades were much less prevalent, especially within senior tranches. In Europe, CLO tranche downgrades were limited to less than 5% of the universe rated BBB/Baa and below.
- **CLO Defaults:** According to Moody's, there were 12 principal impairments in US CLOs in 2020. No tranche defaults or principal impairments have occurred in Europe.

CLO AAA, AA, A New Issue DMs through COVID



CLO BBB, BB, B New Issue DMs through COVID



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Equal-weight/Hold	1736	47%	351	46%	20%	793	47%
Not-Rated/Hold	4	0%	0	0%	0%	1	0%
Underweight/Sell	573	15%	76	10%	13%	230	14%
Total	3,731		765			1673	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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